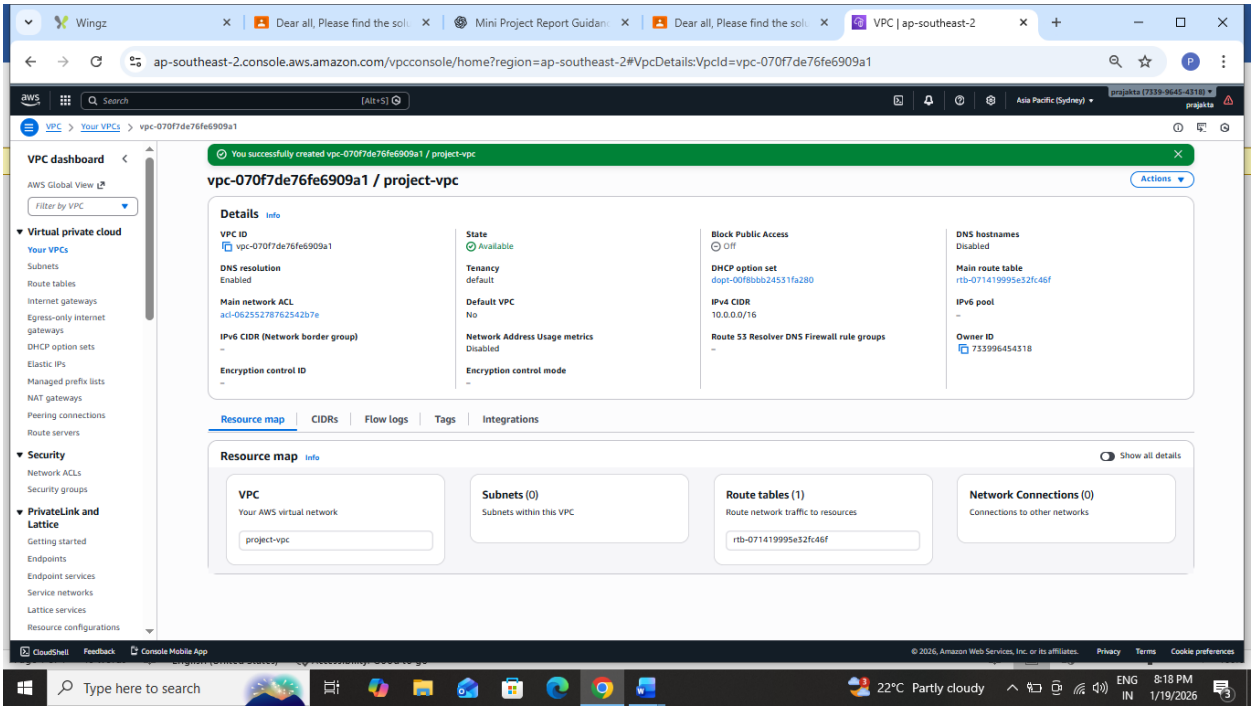
MINI PROJECT -01

TITLE- Secure S3 File Upload Alert System using EC2, SNS, and VPC…

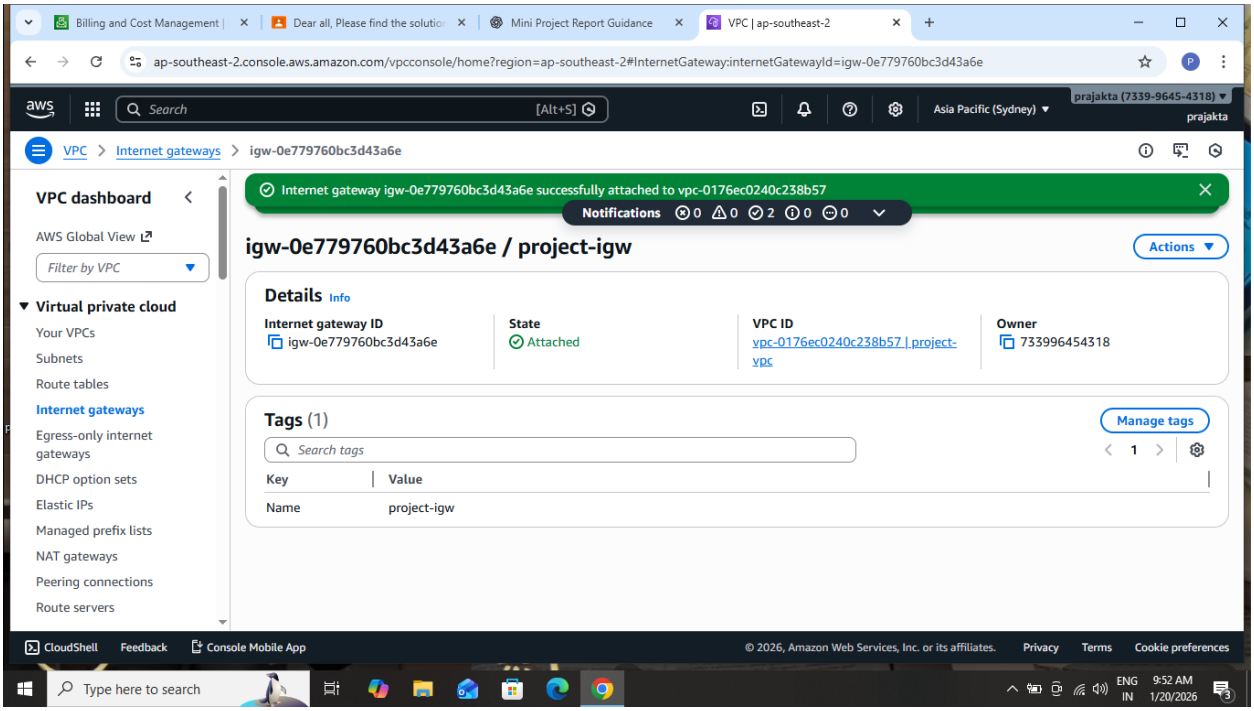
Step 1: Create VPC –



A VPC was created to isolate AWS resources in a private network. This improves security by controlling inbound and outbound traffic using route tables and security groups.

Step 2: created internet gateway and attach to VPC-

An Internet Gateway was attached to the VPC to allow internet access for resources deployed in the public subnet

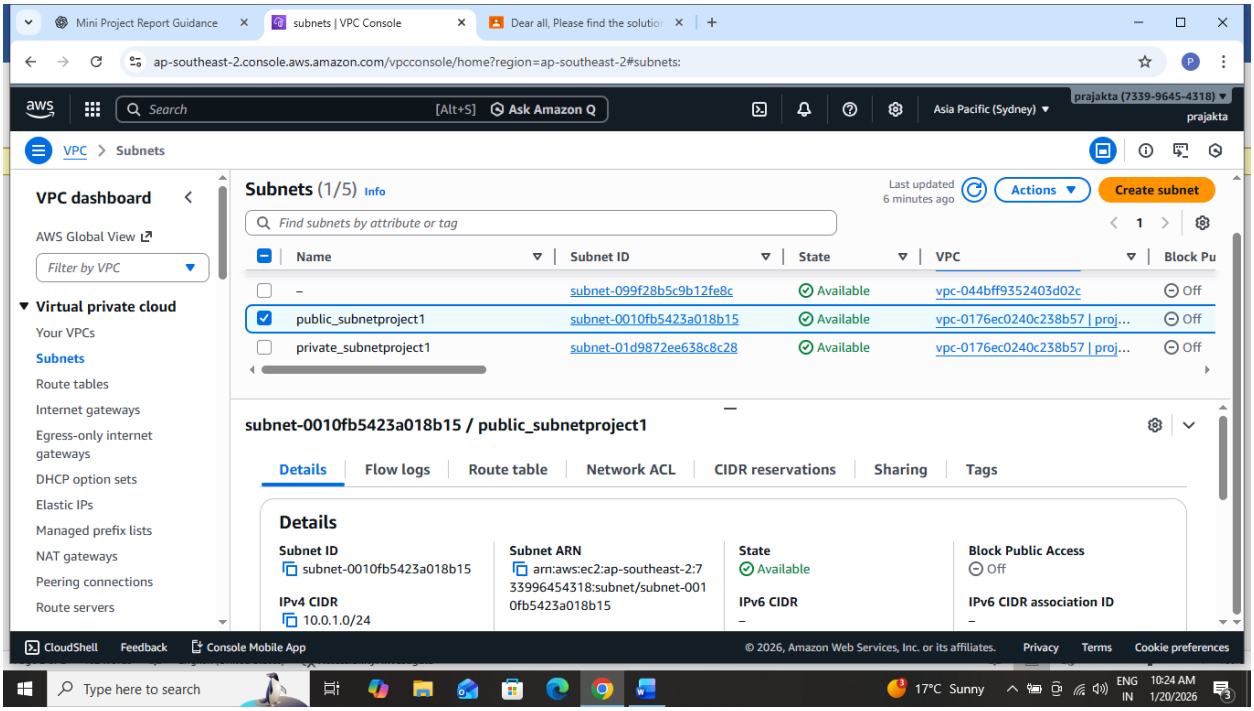


STEP 3- CREATED SUBNET -

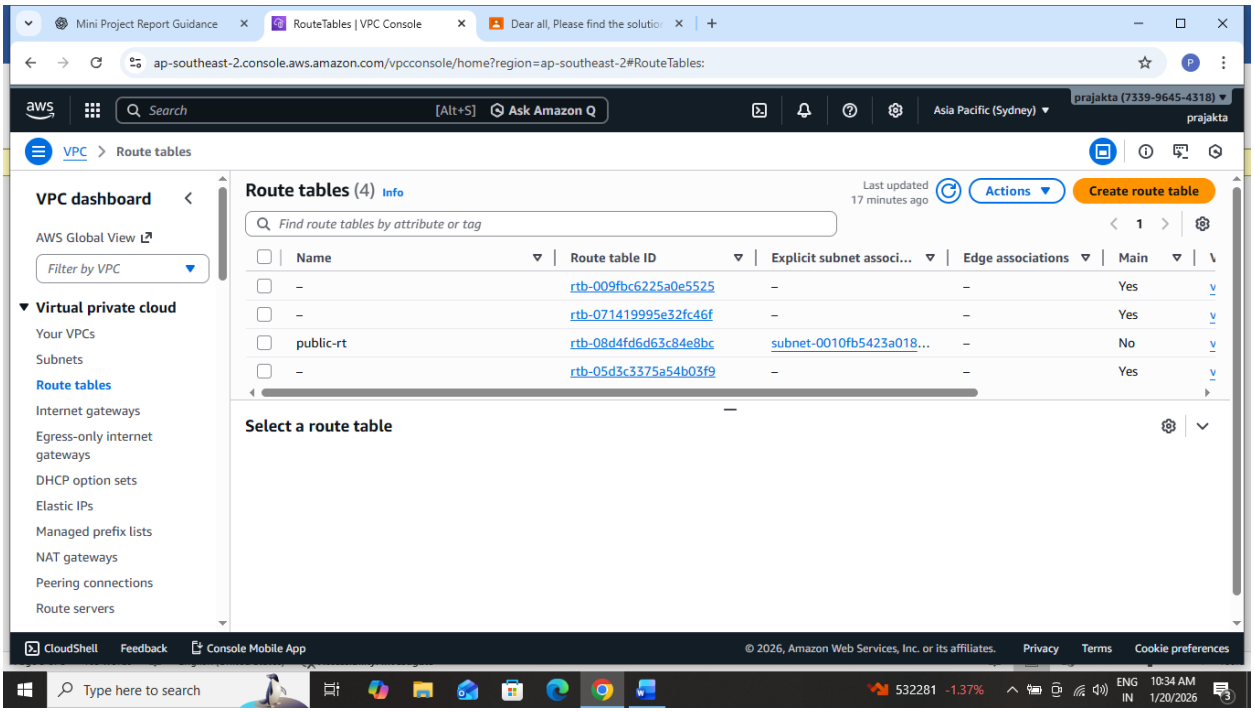
Created 2 subnets public and private.

A subnet is a smaller part of a VPC.

When we create a VPC, we get a **large IP range** (like 10.0.0.0/16).
A subnet divides this large network into **smaller networks** so resources can be organized properly.



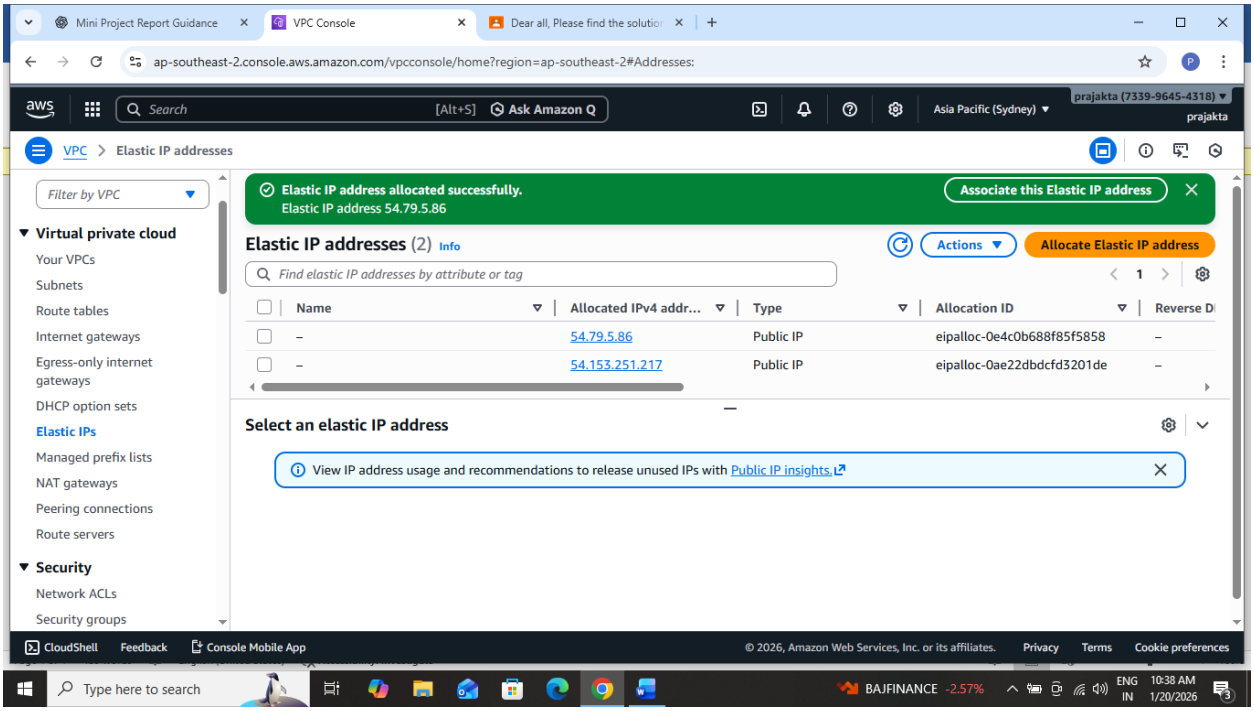
STEP 4- CREATED ROUTTABLE



We need a route table to:

- Control **internet access**
- Decide whether traffic goes to **Internet Gateway**, **NAT Gateway**, or stays inside VPC
- Make a subnet **public or private**

STEP 5- ELASTIC Ips-



What is an Elastic IP?

An **Elastic IP (EIP)** is a **fixed (static) public IP address** provided by AWS.

Normally:

Public IP of EC2 **changes** when instance stops/starts

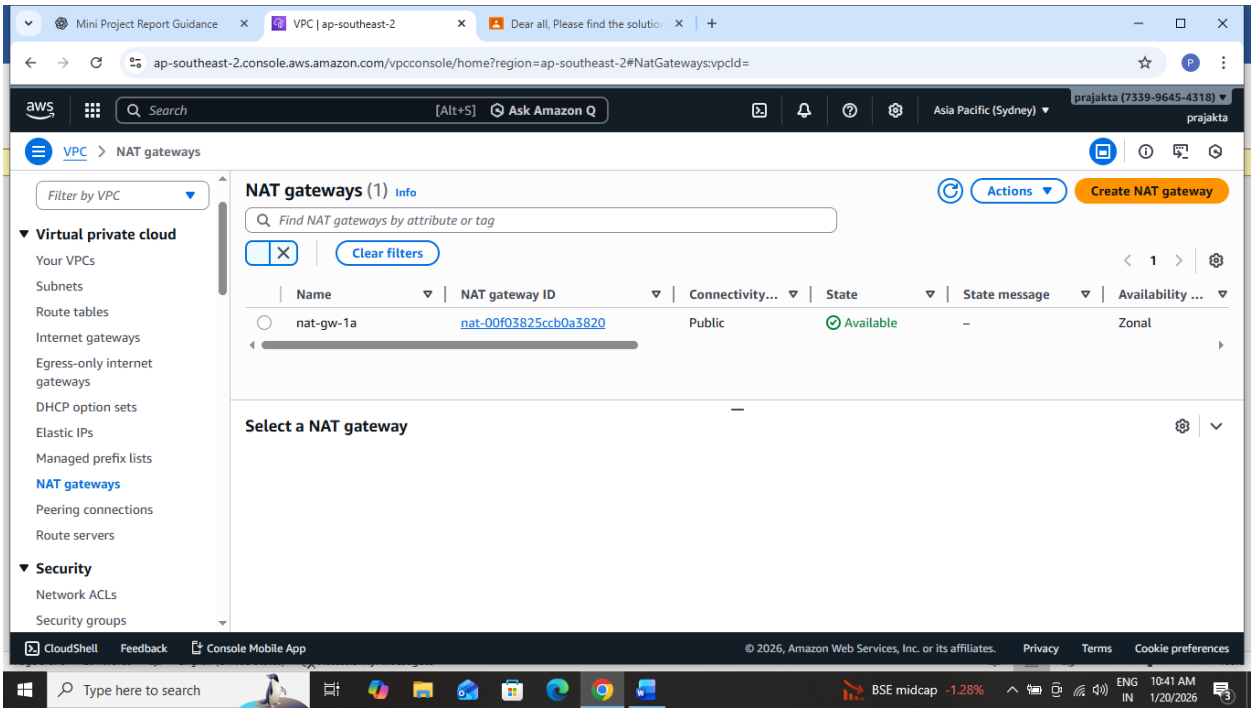
Elastic IP **does NOT change**

What does "Associate Elastic IP" mean?

Associate Elastic IP means:
Attaching a fixed public IP to an EC2 instance

- Your EC2 always has the **same public IP**

STEP 6- CREATE NAT GATEWAY –



What is a NAT Gateway?

NAT means Network Address Translation.

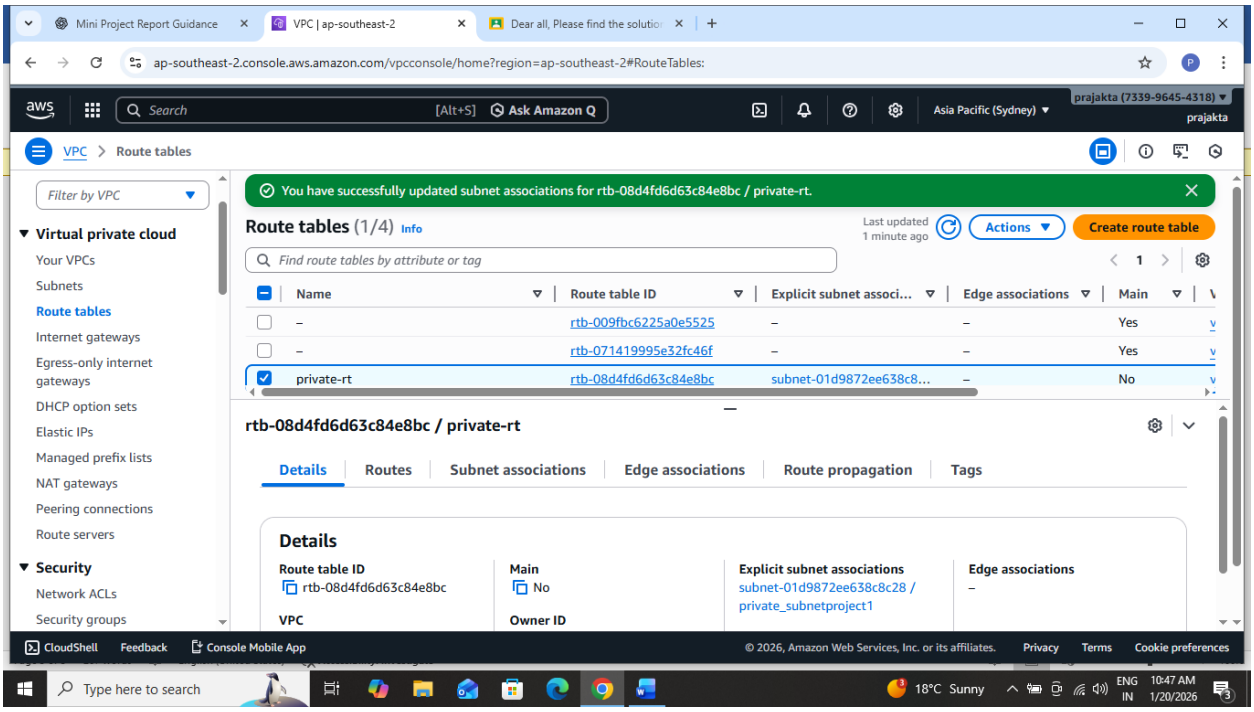
A NAT Gateway allows instances in a private subnet to:
Access the internet
But internet cannot access them directly

Why do we create a NAT Gateway?

We create a NAT Gateway because:

- Private subnet EC2 needs internet (updates, downloads)
- But should remain secure and not publicly accessible

STEP NO 7- create route table



A route table tells AWS where the network traffic should go.

Whenever data moves in or out of a subnet, the route table decides:
Which path the data should take

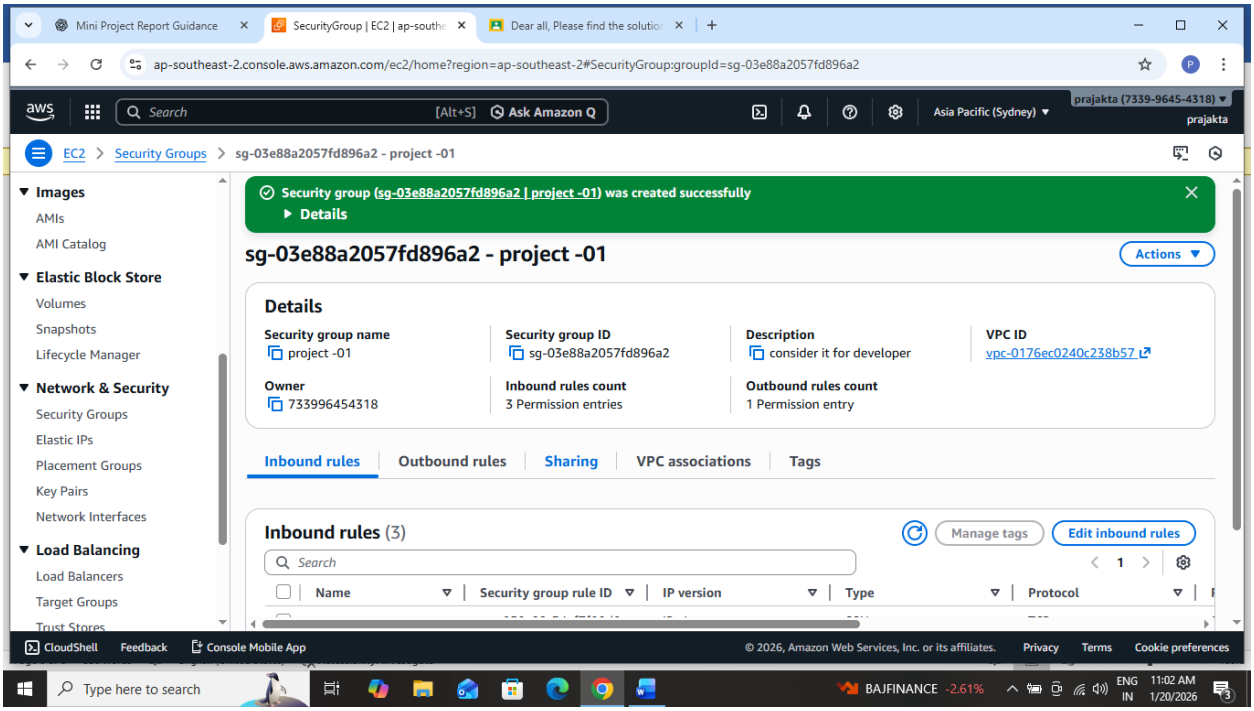
Why do we need a Route Table?

We need a route table to:

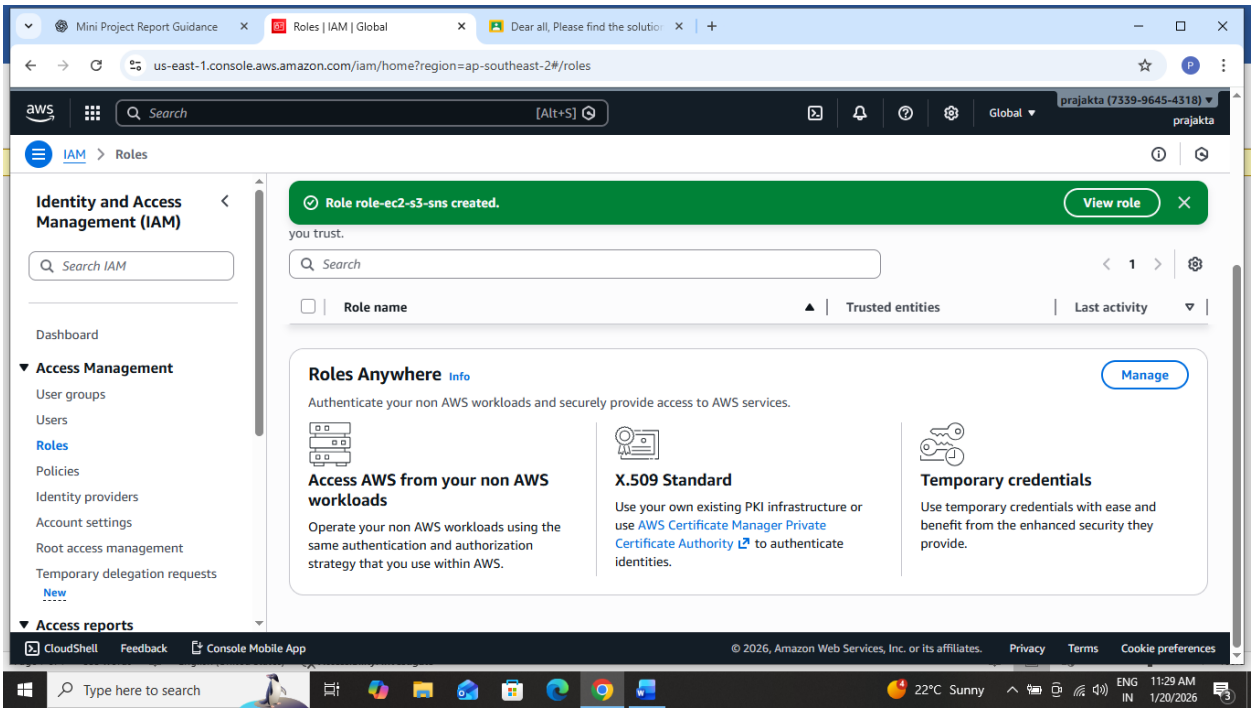
- Connect a subnet to the internet
- Control traffic flow
- Decide if a subnet is public or private

STEP NO 08 –

Created security group-- We create a Security Group to control inbound and outbound traffic to AWS resources like EC2 at the instance level.

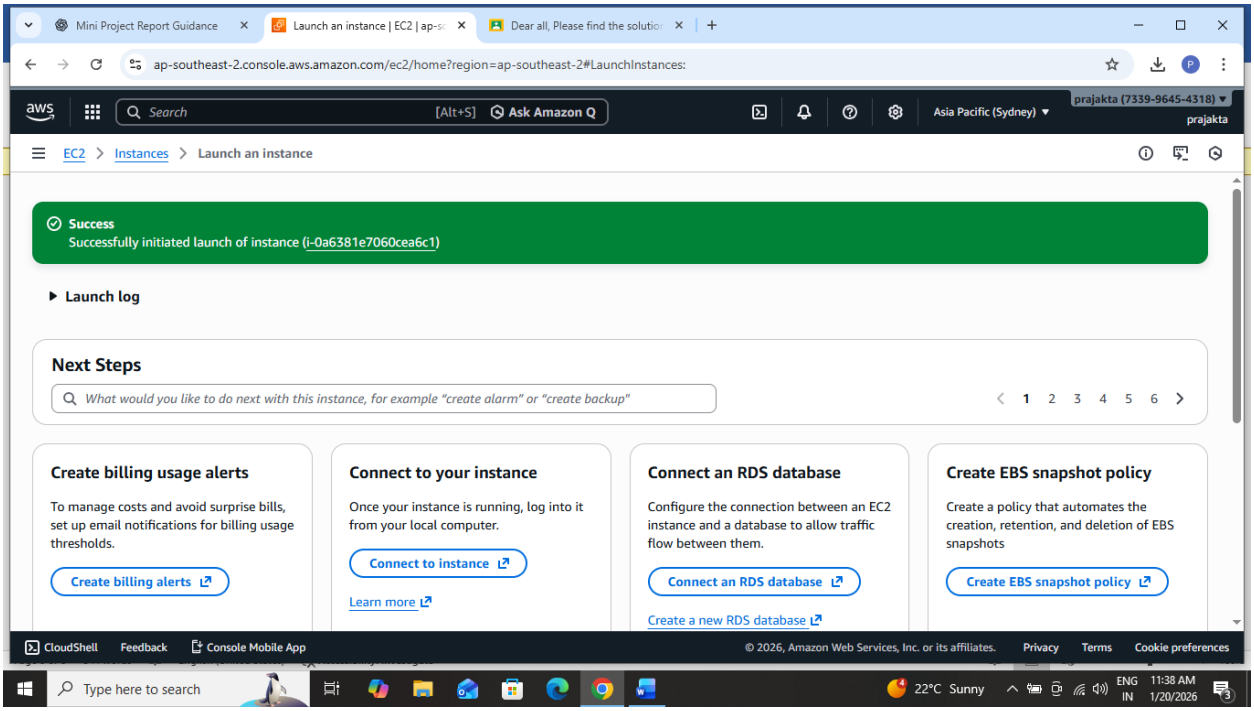


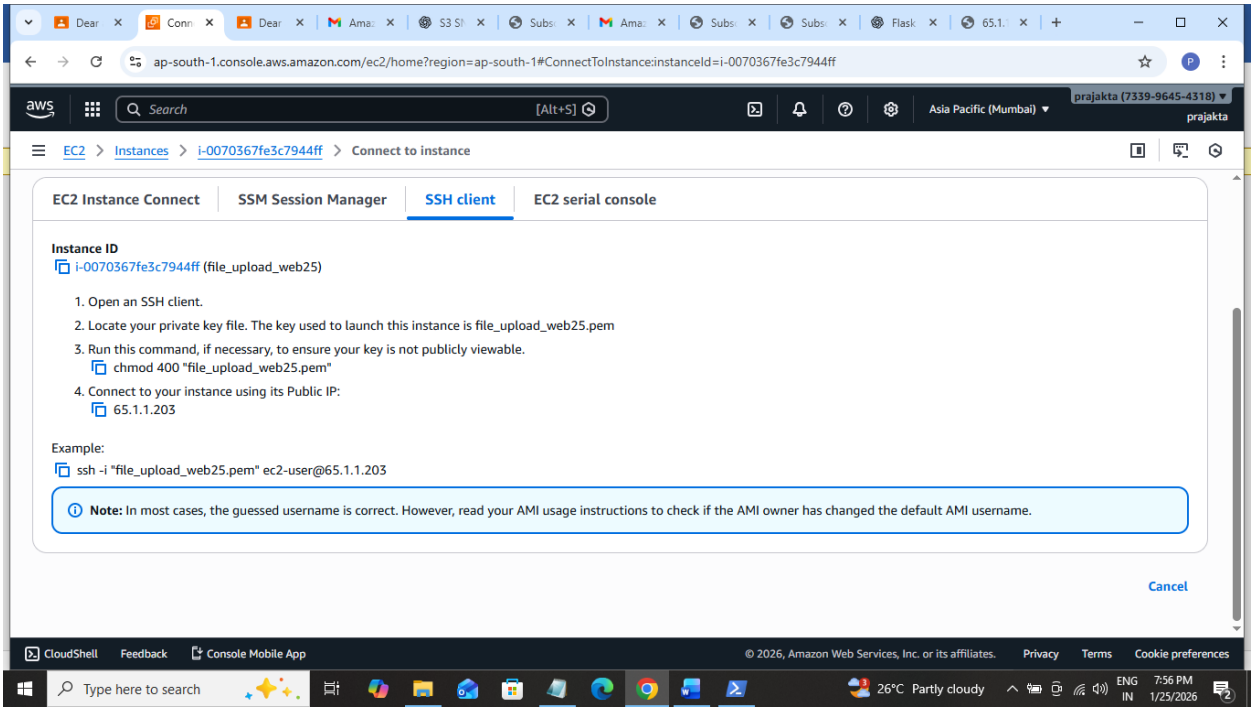
STEP NO 09- CREATE IAM USER AND ADD ROLE –



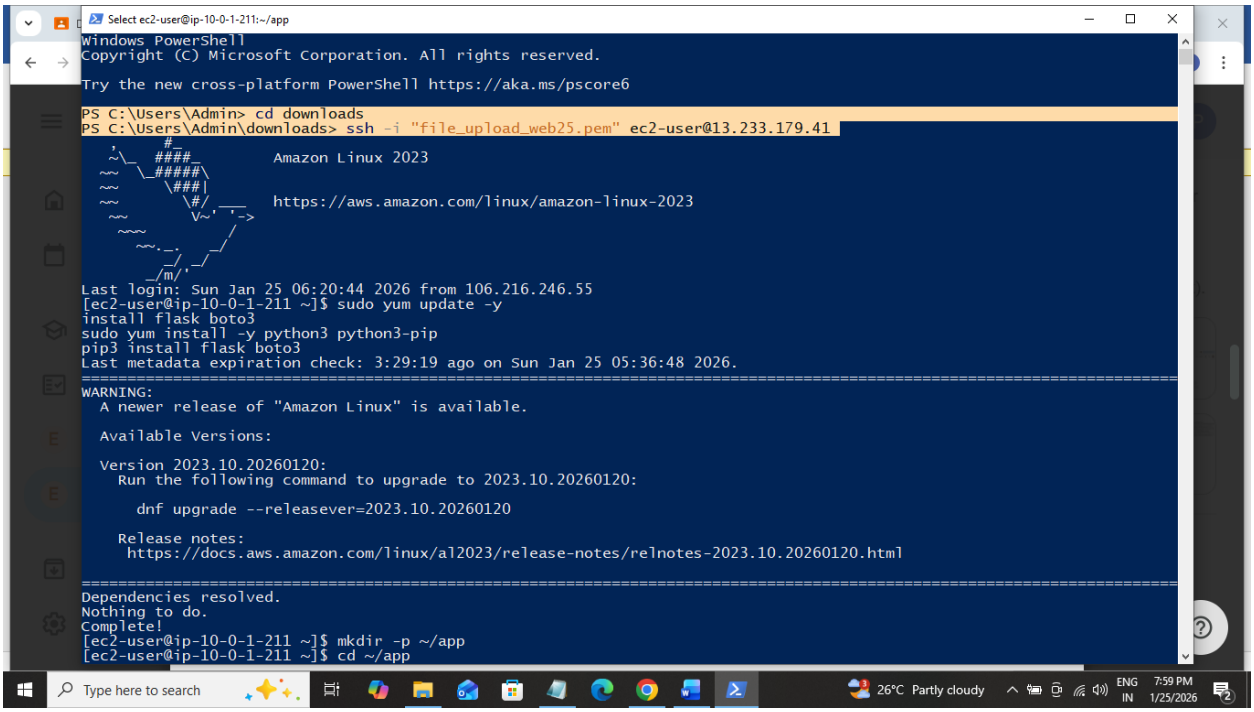
In this project, IAM User is used for secure human access to AWS, while IAM Role is used to grant EC2 permission to upload files to S3 and send notifications via SNS without storing credentials.”

STEP 10- CREATED INSTANCE-





connect instance to SSH.

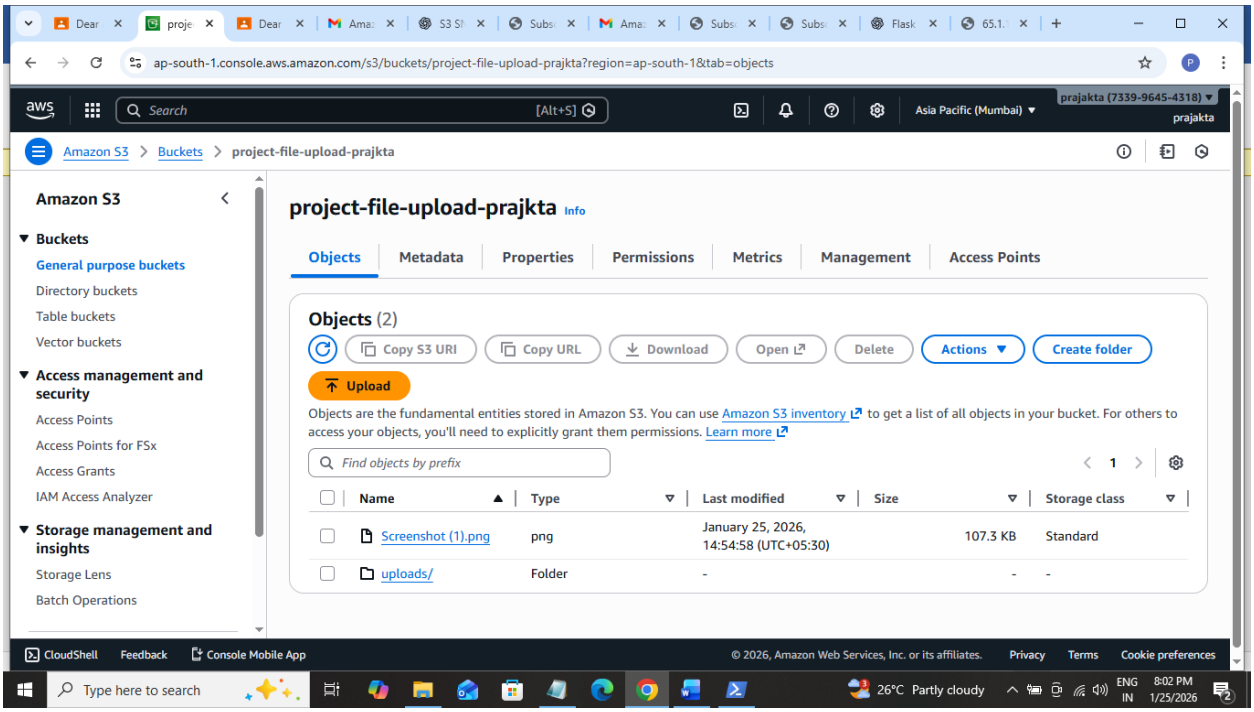


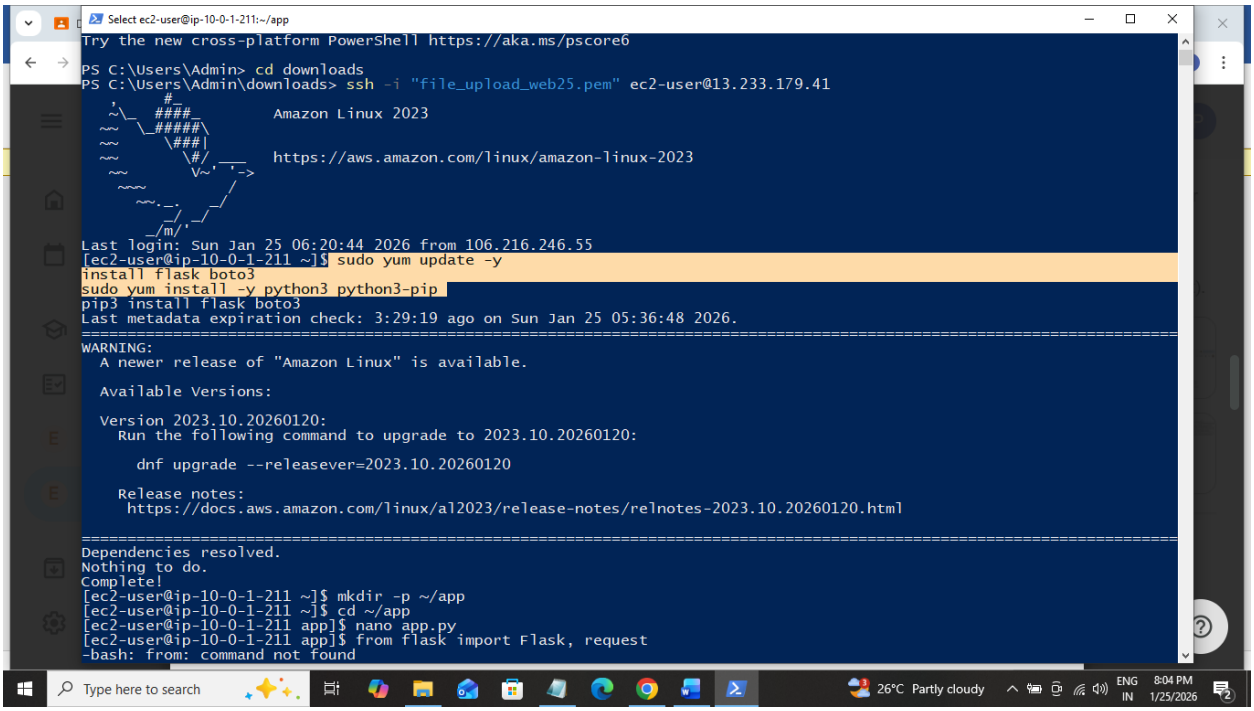
COMMAND USED TO CONNECT THE SERVER ARE-

PS C:\Users\Admin> cd downloads

PS C:\Users\Admin\downloads> ssh -i "file_upload_web25.pem" ec2-user@13.233.179.

STP 11 - CREATE S3 BUCKET---

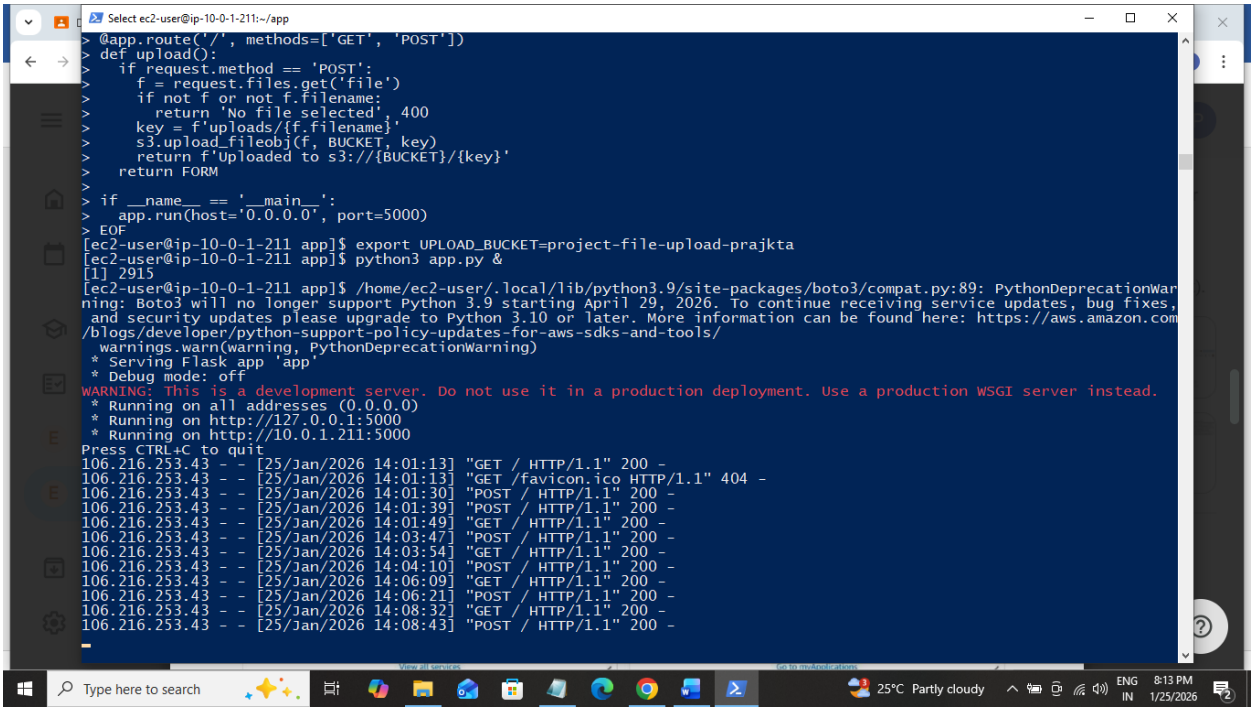




After creating s3 bucket update and install python 3 as shown in above screenshot.

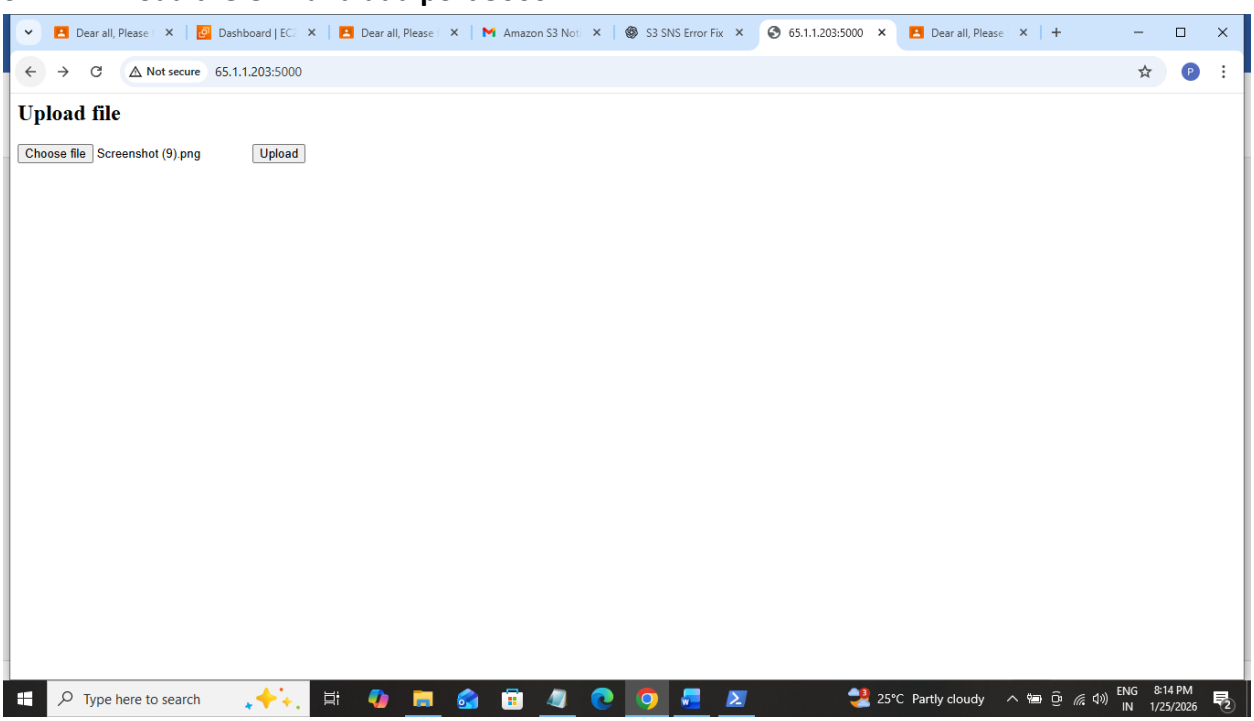
Create an 'app' directory in the ec2-user home and move into it

```
mkdir -p ~/app

cd ~/app

# Import Flask and request object for handling HTTP


cat > app.py << EOF
from flask import Flask, request
import boto3, os

BUCKET = os.environ.get('UPLOAD_BUCKET', 'project-file-upload-prajkta')

app = Flask(__name__)
s3 = boto3.client('s3')

FORM = '''
<h2>Upload file</h2>
<form method=post enctype=multipart/form-data>
  <input type=file name=file>
  <button type=submit>Upload</button>
</form>
'''

@app.route('/', methods=['GET', 'POST'])
def upload():
  if request.method == 'POST':
    f = request.files.get('file')
    if not f or not f.filename:
      return 'No file selected', 400
    key = f'uploads/{f.filename}'
    s3.upload_fileobj(f, BUCKET, key)
    return f'Uploaded to s3://{BUCKET}/{key}'
  return FORM

if __name__ == '__main__':
  app.run(host='0.0.0.0', port=5000)
EOF
```

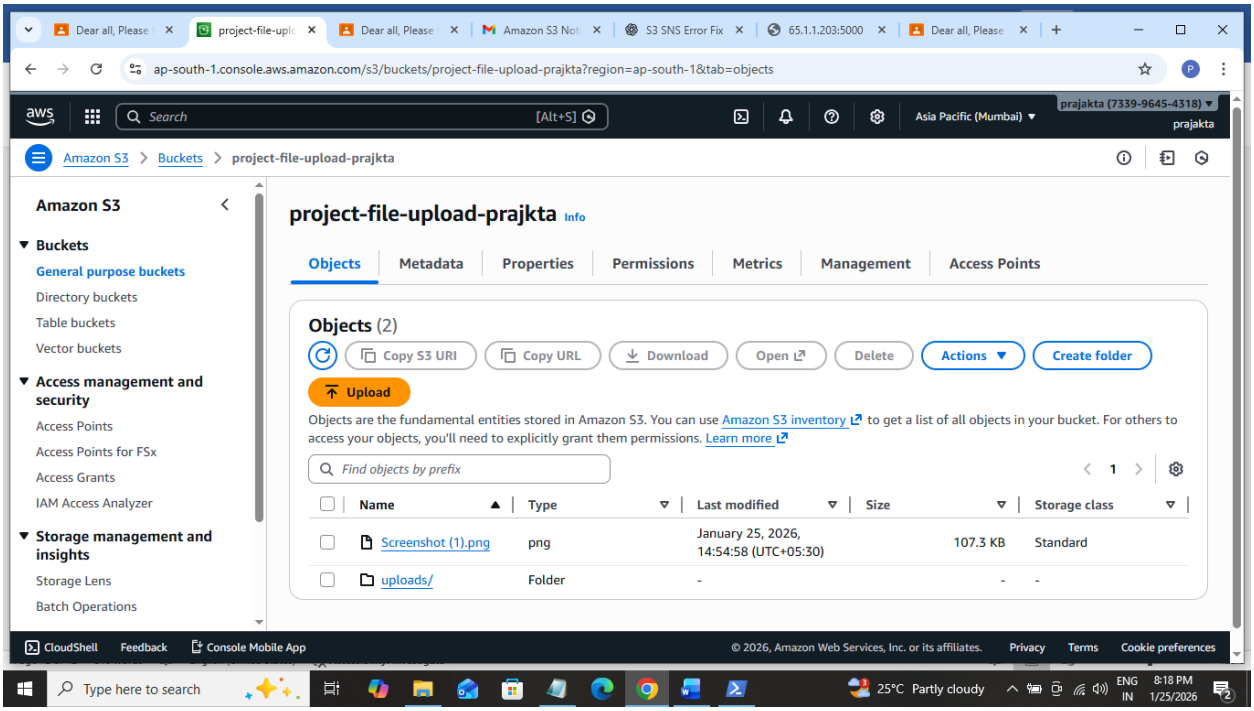


app is created and file is uploaded.

STEP 12- Load the URL and add port 5000 -

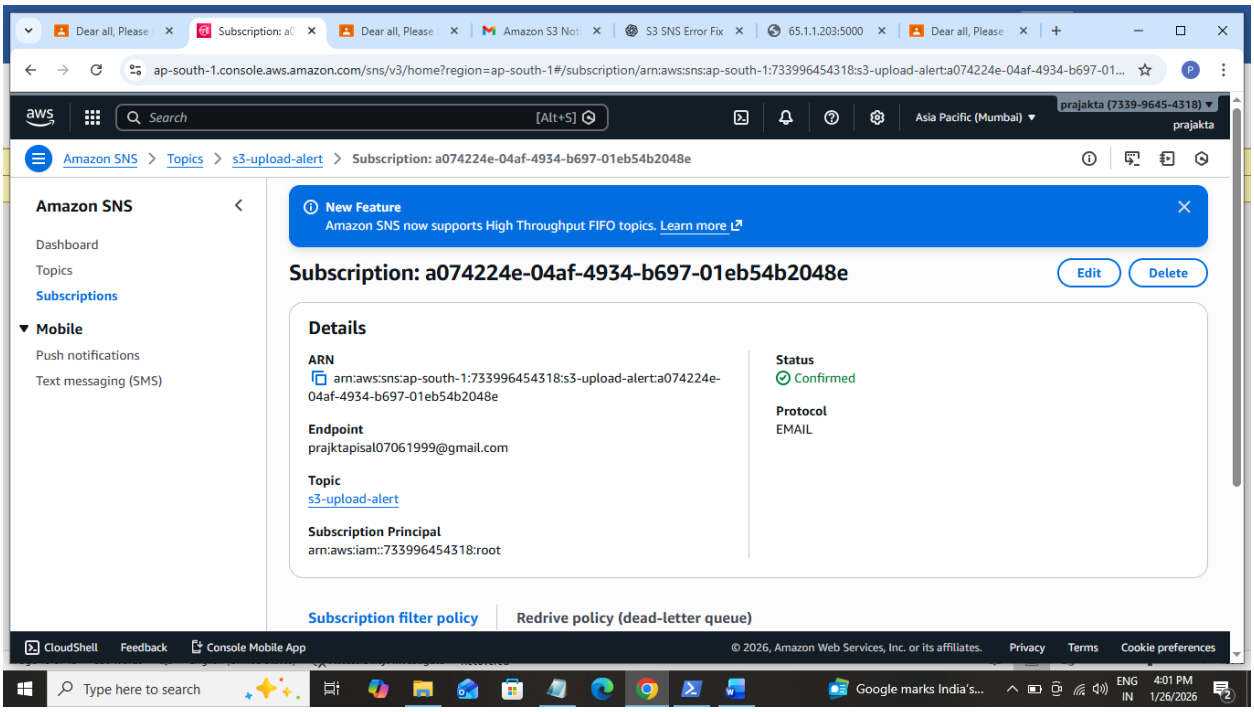


STEP 13- Create Bucket - We created an S3 bucket to store files uploaded by users and to trigger notifications whenever a file is uploaded.

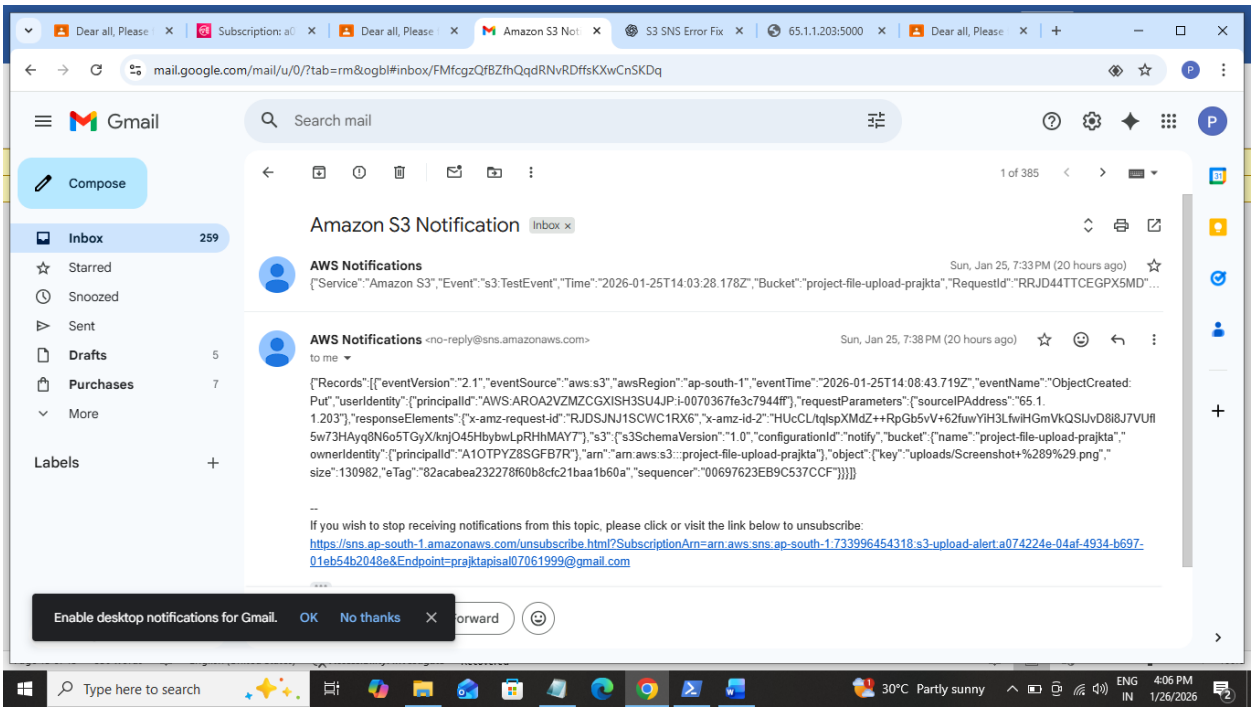


In s3 bucket you can see your uploaded data from web.

STEP 14- CREATE sns (SIMPLE NOTIFICATION SERVICE)— "Amazon SNS is used to send automatic notifications such as email alerts when specific events occur, like file uploads in S3."



STEP 15- The S3 bucket was configured to send notifications to the SNS topic whenever an object is uploaded.



CONCLUSION --- The objective of this project is to design a secure AWS-based system that sends email notifications whenever a file is uploaded to an S3 bucket. The infrastructure is deployed inside a VPC to ensure network security, and notifications are handled using Amazon SNS.

The main goal of this project is to create a system on AWS where:

- **Whenever someone uploads a file to an S3 bucket,**
- **An email alert is automatically sent to the user.**

To make the system safe and secure, all the AWS resources are created inside a VPC, which works like a private network in AWS.

The email alerts are sent using Amazon SNS, which is an AWS service used to send notifications such as emails.

This project shows how to get an email notification when a file is uploaded to S3, using secure AWS services like VPC and SNS.